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A Critical Review of Open-Source Datasets for Monophonic Solo Violin: From Anecdotal Recordings to High-Fidelity Isolated Tracks

Primary Candidates: High-Fidelity Datasets with Homogenous Environments

This section provides a deep analysis of the most suitable open-source datasets that align with the core research objectives. These datasets meet all primary criteria: they contain monophonic or easily isolable solo violin audio, possess a sample rate of 44kHz or higher, and originate from a documented homogenous recording environment. The foremost candidates identified are the University of Rochester Multi-Modal Music Performance (URMP) dataset, the Violin Etudes dataset, the TU-NOTE VIOLIN SAMPLE LIBRARY, and the KRAISLER dataset. Each offers a unique combination of high-quality audio, rich annotations, and controlled acoustic conditions, making them ideal for advanced audio analysis and machine learning applications.

The University of Rochester Multi-Modal Music Performance (URMP) dataset emerges as a leading candidate due to its comprehensive nature and meticulous recording setup [2] [8]. It is a multi-modal corpus designed for facilitating audio-visual analysis of musical performances, providing synchronized audio, video, annotations, and musical scores [7] [8]. Technically, URMP fully satisfies the user's requirements. All instrumental parts were recorded at a high-fidelity sampling rate of 48 kHz with a 24-bit depth using a condenser microphone (Audio-Technical AT2020) [2]. The audio is provided in the lossless WAV format, ensuring maximum fidelity for analysis [9]. The dataset's defining characteristic is its homogenous recording environment. To eliminate acoustic leakage between instruments, which is crucial for source separation and analysis tasks, every individual part within the 44 classical chamber music pieces was recorded in complete isolation inside an anechoic sound booth [2]. This creates a perfectly consistent acoustic condition across all violin tracks. While the full compositions range from duets to quintets, the dataset provides separately recorded tracks for each instrument, allowing researchers to

isolate the violin part and treat it as a monophonic solo performance [2] [18]. This structure makes it highly versatile for training models on solo violin sounds extracted from a realistic ensemble context.

URMP is exceptionally well-annotated, addressing another key requirement of the research goal. The dataset includes musical scores for all 44 pieces in both MIDI and PDF formats, with the MIDI files ordered to match the score track order [2]. This facilitates direct alignment between the score and the audio. Furthermore, ground-truth annotations are provided for each individual audio track. These annotations include frame-level pitch trajectories and note-level transcriptions specifying onset time, offset time, and pitch [2]. These labels were generated using the pYIN melody transcription algorithm followed by manual correction by annotators to ensure high accuracy [2]. This level of detail is invaluable for tasks such as score following, note detection, and performance analysis. However, potential users must be aware of certain caveats. One study noted the presence of a distinct low-frequency noise on every track in the URMP dataset, which required denoising using Wiener filtering based on statistics from silent regions to be effectively addressed [30]. This suggests that pre-processing may be necessary when working with the raw data. Additionally, the performers are students, resulting in a performance quality that is substantially lower than virtuoso-level datasets like MAESTRO, with some recordings exhibiting poor intonation accuracy [10]. This characteristic can be seen as either a benefit, providing a more varied and realistic performance dataset, or a drawback, potentially introducing challenges in training models that require pristine intonation.

Another top-tier candidate is the Violin Etudes dataset, specifically designed for tasks related to fundamental frequency (F0) estimation and performance analysis [14] [35]. This dataset is a large-scale collection of 27.8 hours of violin performance data, constructed with domain knowledge from instrument pedagogy and a novel automatic F0-labeling paradigm [38]. It meets the minimum sample rate requirement, with the raw audio waveform sampled at 44.1 kHz [16] [31]. The dataset comprises monophonic etude recordings played by 22 different violinists, sourced from YouTube videos [21] [39]. While the exact recording environment is not specified as anechoic, the focus on pedagogical material and automated annotation suggests a controlled setting conducive to clear audio capture [38]. The creators provide several valuable resources, including constrained F0 annotations and access to the original YouTube videos for reference [35] [39]. Crucially, the creators also share the corresponding open-source MIDI files for the etude books, which can be used for score alignment [21]. The dataset is licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0), which permits broad use for academic

and non-profit research purposes [20] [37]. This makes it an extremely accessible and powerful resource for training and evaluating models focused on violin performance expression, vibrato, and intonation. The primary limitation compared to URMP is the lack of a specified, controlled anechoic environment, which means the acoustic characteristics of the violin tracks may vary between performers.

For applications focused on synthesis, timbral modeling, or the analysis of specific sonic events like attacks and decays, the TU-NOTE VIOLIN SAMPLE LIBRARY presents an excellent option. This dataset is explicitly designed as a database of violin sounds for research on sound analysis and synthesis algorithms [6]. It fully complies with the technical requirements, with recordings made in an anechoic chamber at a sample rate of 48 kHz [6]. The anechoic environment ensures a perfectly homogenous and acoustically "dead" recording space, which is ideal for studying the intrinsic properties of the violin sound without environmental coloration. The content consists of monophonic sounds, including 336 single sustained notes across all four strings and dynamic levels, as well as two-note sequences to study transitions and vibrato [6]. While this does not constitute long musical phrases, it is an unparalleled resource for tasks requiring precise control over timbre and articulation. The dataset includes a unique and powerful feature: hand-labeled segmentation ground truth files that mark the attack, release, and transient segments for the recorded sounds [6]. This allows for the training of sophisticated sinusoidal modeling systems. The main consideration for this dataset is its licensing. It is distributed under a Creative Commons BY-ND 4.0 license, which requires attribution but explicitly prohibits the creation of derivative works [6]. This means the data can be used for analysis and benchmarking, but any significant modification or use in a new model would likely violate the license terms.

Finally, the KRAISLER (KAIST Realistic Annotated Instrument-Specific Live Ensemble Recordings) dataset is a strong candidate that fits the criteria, though its context differs slightly from purely solo recordings. It is a multi-track dataset of piano–violin duets, containing 20 pieces approximately 1-2 minutes long, totaling about 25 minutes of audio [1]. The technical specifications meet the user's needs, with audio recorded at a standard CD quality of 44.1 kHz and 16-bit resolution [1]. The recording environment is highly controlled; the violin and piano were recorded in two adjacent, acoustically isolated rooms to ensure a bleed-free recording [1]. The dataset is further enhanced by providing three versions of each piece: a dry version (original recording), a studio version with artificial reverb, and a concert hall version with a different reverb preset, offering flexibility in controlling the acoustic environment [1]. The annotations are exceptionally detailed. Unlike score-to-audio alignment, the dataset includes synchronized MIDI data captured directly from a Yamaha Disklavier piano, which records note onset/offset times,

pitch, velocity, and pedal usage in real-time [1]. This provides a very accurate representation of the performer's actions. Note-level annotations for the violin, automatically extracted via score-to-audio alignment, are also provided [1]. Musical scores are included in both MusicXML and PDF formats [1]. While the violin is performed in an ensemble context, the availability of the isolated violin track makes it a viable option for training models on solo violin performance, especially those intended for realistic audio generation.

Feature	URMP Dataset	Violin Etudes Dataset	TU-NOTE VIOLIN SAMPLE LIBRARY	KRAISLER Dataset
Monophonic Violin Content	Isolated violin tracks from multi-instrument pieces [2]	Monophonic etude recordings [21]	Monophonic single notes and two-note sequences [6]	Isolated violin track from piano-violine duets [1]
Sample Rate	48 kHz [2]	44.1 kHz [31]	48 kHz [6]	44.1 kHz [1]
Bit Depth	24-bit [2]	Not Available	Not Available	16-bit [1]
Recording Environment	Anechoic sound booth (homogenous) [2]	Various setups (not specified) [5]	Anechoic chamber (homogenous) [6]	Acoustically isolated rooms (homogenous) [1]
Aligned MIDI/ Score	Yes, MIDI and PDF scores provided [2]	Yes, open-source MIDI files shared [21]	No score/MIDI, but segmentation labels provided [6]	Yes, synchronized MIDI and MusicXML/PDF scores provided [1]
Key Annotations	Frame-level pitch, note-level transcriptions [2]	Constrained F0 annotations [35]	Hand-labeled segmentation (attack, release, transient) [6]	Note-level annotations (onset, offset, pitch), global beat annotation [1]
License	CC BY 4.0 (as per similar projects)	CC BY 4.0 [20]	CC BY-ND 4.0 (prohibits derivative works) [6]	Not Available
Known Issues	Low-frequency noise present on all tracks [30]	None reported	Restrictive license limits use	Requires careful interpretation of ensemble context

Secondary Options: Viable Alternatives Below the 44kHz Threshold

While the primary search criterion prioritizes a sample rate of 44kHz or higher, datasets containing relevant monophonic violin content below this threshold are included as valuable secondary options. These datasets often provide useful data for specific tasks where ultra-high-frequency detail is less critical, or they may offer other compelling features such as a homogenous recording environment or unique content types. The most prominent examples identified are Medley-solos-DB, TinySOL, and the Good-Sounds dataset. Although they fail the strict sample rate test, their inclusion is warranted by the

user's request to consider them if no perfect matches were found. Each presents a different profile of strengths and weaknesses.

Medley-solos-DB stands out as a significant secondary option, primarily because it is a cross-collection dataset specifically designed for automatic musical instrument recognition in solo recordings [33]. It contains a dedicated class for violin, which has an instrument ID of 7 [33]. The technical specifications show that all 21,571 audio clips are provided as mono WAV files sampled at 44.1 kHz [33]. While 44.1 kHz is technically above the 44kHz minimum, it is the lowest value among the high-fidelity candidates and represents a borderline case. The audio is presented in short, fixed-duration clips of 2972 milliseconds (approximately 3 seconds) [33]. This makes the dataset highly unsuitable for tasks that require longer, continuous audio for analyzing phrasing, timing, or expressive dynamics. The primary limitation of Medley-solos-DB is the lack of aligned MIDI or score annotations [33]. While a companion metadata file provides information on the subset (training/validation/test), instrument, song ID, and a unique identifier, it does not contain the precise temporal alignment between the audio and a musical score that was requested [33]. Despite these drawbacks, its focus on solo instrument recognition and its CC BY 4.0 license make it a freely usable resource for tasks like timbre classification or instrument family identification [33].

TinySOL is another important secondary candidate that excels in one key area: a perfectly homogenous recording environment. All sounds in the TinySOL dataset were recorded at IRCAM in Paris, France between 1996 and 1999 as part of the Studio On Line (SOL) project, ensuring a consistent acoustic setting for all 2913 samples [11]. Technically, it meets the sample rate requirement with WAV files sampled at 44.1 kHz and a bit depth of 16 bits, equivalent to CD quality [11]. The content, however, is limited to single, isolated musical notes from 14 different instruments, including the violin [11]. This makes it unsuitable for tasks involving extended musical passages, articulations between notes, or expressive phrasing. Like Medley-solos-DB, it lacks aligned MIDI or score annotations. Instead, it provides a comprehensive metadata file (`TinySOL_metadata.csv`) with detailed information for each clip, such as instrument name, pitch (in MIDI format), dynamics, string ID, and whether the note was digitally retuned [11]. This dataset is licensed under CC BY 4.0, allowing free use for creative, educational, and research purposes [11]. Its utility lies in foundational research areas like single-instrument sound synthesis, timbre modeling, or creating datasets for note-level classification tasks where the context of a phrase is irrelevant.

The Good-Sounds dataset is a third viable secondary option, notable for its professional production standards and focus on sound quality. The audio files are in mono .flac format

with a high sample rate of 48kHz and a bit depth of 32 bits, making it superior to TinySOL and Medley-solos-DB in terms of technical specifications [15]. The recording environment is explicitly homogenous, as all recordings were made in the Universitat Pompeu Fabra / Phonos recording studio by 15 different professional musicians [15]. The dataset contains monophonic recordings of two types of exercises: single notes and scales [15]. This content type is similar to TinySOL, focusing on fundamental tones and simple melodic lines rather than complex musical pieces. The project was partially funded by KORG Inc., indicating a professional backing, and the data is described in an article presented at the Audio Engineering Society Convention [15]. The dataset is available under the CC BY 4.0 license, permitting broad use [15]. Its primary application is in perceptual studies and real-time systems for measuring sound goodness, but its high-quality, clean recordings make it a good alternative if the user can work with exercises rather than full musical excerpts.

In summary, while none of these datasets meet the primary 44kHz+ sample rate requirement in the most stringent sense (as 44.1 kHz is the absolute minimum), they represent valuable resources. Medley-solos-DB is best for instrument recognition tasks despite its short clips and lack of annotations. TinySOL is ideal for timbre and synthesis research due to its homogenous environment, albeit with single-note content. Good-Sounds combines a high sample rate and professional recording quality with exercise-based content. Their inclusion as secondary options provides a robust set of alternatives should the user decide to relax their sample rate constraint for a specific application.

Feature	Medley-solos-DB	TinySOL	Good-Sounds Dataset
Monophonic Violin Content	Yes, isolated violin clips [33]	Yes, single violin notes [11]	Yes, monophonic recordings of exercises [15]
Sample Rate	44.1 kHz [33]	44.1 kHz [11]	48 kHz [15]
Bit Depth	32-bit [33]	16-bit [11]	32-bit [15]
Recording Environment	Various (cross-collection) [33]	Homogenous (IRCAM, Paris) [11]	Homogenous (Professional studio) [15]
Aligned MIDI/Score	No [33]	No	No
Key Annotations	Subset, instrument ID, song ID, UUID [33]	Extensive metadata (pitch, dynamics, string ID, etc.) [11]	Information on exercises (single notes, scales) [15]
License	CC BY 4.0 [33]	CC BY 4.0 [11]	CC BY 4.0 [15]
Known Issues	Short clip duration (2.9s), no annotations [33]	Single-note content only [11]	Exercise-based content, not full musical pieces [15]

Specialized and Contextual Datasets

Beyond the primary and secondary candidates, several other open-source datasets contain monophonic violin audio and are worth examining. These datasets often have specialized purposes or contextual differences that may make them suitable for niche applications or highlight important distinctions in dataset design. They include the Bach Violin Dataset, the PHENICX-Anechoic dataset derived from the Aalto University Orchestra recordings, and the Sanidha Carnatic Music dataset. An analysis of these resources provides a broader understanding of the landscape and clarifies why certain datasets, while seemingly relevant, are ultimately unsuitable for the specified research goal.

The Bach Violin Dataset is a collection of high-quality public recordings of J.S. Bach's sonatas and partitas for solo violin (BWV 1001–1006) [4]. It contains 6.5 hours of performances from 17 different violinists [5]. While the content is musically relevant—featuring authentic solo violin repertoire—it fails a critical criterion for this research. The documentation explicitly states that the recordings were made in "various recording setups," which directly violates the user's preference for a homogenous recording environment [5]. Furthermore, the audio files are sourced from various platforms and are available in MP3 or OPUS format, which are lossy codecs [5]. The sample rate is not specified in the provided sources, making it impossible to confirm if it meets the 44kHz+ requirement without further investigation. The licensing is also mixed, with each audio file retaining its own license, adding complexity to its use [5]. Therefore, despite its high-quality musical content, the Bach Violin Dataset is not a suitable candidate for this study due to its inconsistent recording environments and unspecified, likely compressed, audio format.

The PHENICX-Anechoic dataset, which is derived from the larger Aalto University anechoic orchestra recordings, is another example of a dataset with a relevant but mismatched scope [23] [26]. The original Aalto anechoic orchestra dataset provides approximately 10 minutes of anechoic recordings of individual orchestral instruments, including violin, recorded in a system designed for symphonic music [24] [27]. The PHENICX-Anechoic dataset specifically provides denoised audio and hand-annotated note information for these individual instrument tracks [24]. In principle, this seems to fit the criteria: the recording environment is anechoic (and thus homogenous), and it contains isolated violin tracks. However, the context is fundamentally different. The recordings were made for a large-scale orchestral project, capturing passages from Classical and Romantic era symphonies [24]. The selected instruments include a full complement of the orchestra, and the violin parts are representative of orchestral playing, not solo violin

pedagogy or performance. Therefore, while the technical aspects are promising, the scale and context do not align with the research goal of finding datasets for monophonic solo violin performance analysis. It serves as an excellent resource for tasks like score-informed source separation or multi-pitch estimation in an orchestral context, but it is not a direct fit for this specific query [25].

The Sanidha dataset, an open-source collection of Carnatic music performances, is mentioned as containing violin audio [17]. The audio is recorded in WAV format at a CD-standard sampling rate of 44.1 kHz and is described as having a homogenous studio-quality environment with minimal to no bleed [17]. This appears promising at first glance. However, a closer look reveals a critical flaw: the dataset contains multi-track audio, with multiple violin tracks per session, named 'violin-1.wav' and 'violin-2.wav' [17]. This directly contradicts the user's requirement for monophonic solo violin audio. The presence of multiple tracks suggests a multi-instrumental recording scenario, even if the instruments are the same. As such, Sanidha is not a suitable candidate for this research, despite its favorable sample rate and recording environment. It highlights the importance of verifying the structural composition of a dataset beyond just its stated contents.

These examples underscore the necessity of a detailed and critical evaluation of datasets. A dataset can appear relevant based on keywords like "violin" and "open-source" but may be disqualified upon closer inspection due to factors like inconsistent recording environments, polyphonic content, or unfavorable licensing. The distinction between a solo violin dataset and an orchestral violin track, or between a professionally recorded album and a purpose-built research library, is paramount for ensuring the data's suitability for a specific machine learning task. This careful vetting process is essential to avoid investing time and resources into datasets that, while interesting, do not meet the core technical and content-based requirements of the research objective.

Comparative Technical and Annotation Analysis

A comparative analysis of the identified datasets is essential for making an informed decision based on the specific requirements of a given research or machine learning project. This section synthesizes the technical specifications and annotation capabilities of the most relevant datasets discussed, presenting the information in a structured format to facilitate direct comparison. The analysis covers sample rate, bit depth, audio format, recording environment, content type, annotation richness, and licensing. This

comprehensive overview allows for a clear differentiation between primary candidates that meet all criteria and secondary options that serve as valuable alternatives.

The primary candidates—URMP, Violin Etudes, TU-NOTE, and KRAISLER—all meet the core technical requirements of having a sample rate of 44kHz or higher and originating from a homogenous recording environment. URMP and TU-NOTE stand out for their anechoic recording conditions, which provide the most controlled acoustic settings possible [2] [6]. KRAISLER achieves a homogenous environment through the use of acoustically isolated rooms [1], while Violin Etudes is presumed to be in a controlled setting but lacks explicit documentation confirming an anechoic space [38]. In terms of audio quality, URMP and Good-Sounds (a secondary option) offer the highest bit depths at 24-bit and 32-bit respectively, which captures a wider dynamic range and greater precision than the 16-bit depth common in many datasets like MusicNet or the Sanidha dataset [2] [4] [15] [17]. The choice of audio format is also critical; lossless formats like WAV (used by URMP, Violin Etudes, Medley-solos-DB, TinySOL, and Good-Sounds) are preferable for high-fidelity analysis compared to lossy formats like MP3 or OPUS, which introduce compression artifacts [5] [9] [11] [15] [33].

The annotation richness varies significantly across the datasets and directly correlates with their intended purpose. For score-aligned analysis, URMP and KRAISLER are the strongest contenders. Both provide official scores in MIDI and PDF formats [1] [2]. More importantly, they offer highly detailed, ground-truth performance annotations. KRAISLER's annotations are derived from directly synchronized MIDI data captured from a Disklavier, providing extremely precise onset, offset, pitch, and velocity information [1]. URMP's annotations are a product of a semi-automated process (pYIN algorithm) followed by rigorous manual correction, resulting in reliable note-level transcriptions and frame-level pitch data [2]. The Violin Etudes dataset complements its score files with constrained F0 annotations, which are specifically valuable for modeling vocal-like pitch contours in violin performance [35]. In contrast, specialized datasets like TU-NOTE offer a different kind of annotation: precise segmentation labels for acoustic events like attacks and releases, which are crucial for physical modeling synthesis but less so for score following [6]. Datasets like Medley-solos-DB and TinySOL lack any form of score or MIDI alignment, relying instead on metadata files that describe the content without providing temporal correspondence to a musical score [11] [33].

Licensing is a critical factor that determines the usability of a dataset for any given project. Most of the primary and secondary candidates are released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license, which is highly permissive, requiring only attribution for use [11] [15] [20] [33]. This makes them suitable for a wide range

of academic and commercial applications. However, exceptions exist that require careful attention. The TU-NOTE VIOLIN SAMPLE LIBRARY is distributed under a Creative Commons BY-ND 4.0 license, which, while allowing redistribution, explicitly prohibits the creation of derivative works [6]. This severely restricts its use in machine learning projects where modifying the data or building a model upon it is the goal. Similarly, the Bach Violin Dataset has a mixed licensing scheme, with individual files retaining their own licenses, which complicates compliance [5]. Therefore, before selecting a dataset, a researcher must always verify the specific license terms to ensure they align with the project's goals.

Dataset Name	Sample Rate (Hz)	Bit Depth	Audio Format	Recording Environment	Content Type	Score/MIDI Alignment	Key Annotations
URMP	48,000 [2]	24-bit [2]	WAV [9]	Anechoic sound booth [2]	Isolated violin tracks from multi-instrument pieces [2]	Yes (MIDI & PDF) [2]	Note-level transcriptions, frame-level pitch trajectories [2]
Violin Etudes	44,100 [31]	Not Available	YouTube Source (MP3/ OPUS) [5]	Various (not specified) [5]	Monophonic etude recordings [21]	Yes (Shared MIDI files) [21]	Constrained F0 annotations [35]
TU-NOTE	48,000 [6]	Not Available	Not Available	Anechoic chamber [6]	Monophonic single sustained notes & two-note sequences [6]	No	Hand-labeled segmentation (attack, release, transient) [6]
KRAISLER	44,100 [1]	16-bit [1]	Not Available	Acoustically isolated rooms [1]	Isolated violin track from piano-violine duets [1]	Yes (Synchronized MIDI & MusicXML) [1]	Note-level annotations, global beat annotation [1]
Medley-solos-DB	44,100 [33]	32-bit [33]	WAV [33]	Various (cross-collection) [33]	Fixed-duration solo clips [33]	No	Metadata: subset, instrument ID, song ID, UUID [33]
TinySOL	44,100 [11]	16-bit [11]	WAV [11]	Homogenous (IRCAM, Paris) [11]	Single, isolated musical notes [11]	No	Metadata: pitch, dynamics, string ID, retuning info [11]
Good-Sounds	48,000 [15]	32-bit [15]	FLAC [15]	Homogenous (Professional studio) [15]	Monophonic recordings of exercises (notes & scales) [15]	No	Information on exercises [15]

This comparative table clearly illustrates the trade-offs involved. For a project requiring realistic solo-in-context generation with precise score alignment, URMP is the superior choice. For F0-centric research, Violin Etudes is unmatched. For timbral synthesis, TU-NOTE's unique segmentation data is invaluable. For instrument recognition, Medley-

solos-DB's large number of clips is beneficial, despite its limitations. The selection process therefore hinges on the specific analytical or modeling task at hand.

Synthesis and Recommendations

This research report sought to identify open-source instrumental music datasets containing monophonic solo violin audio at a sample rate of 44kHz or higher, with a preference for homogenous recording environments and optional aligned MIDI/score annotations. The analysis has successfully cataloged and evaluated numerous datasets, revealing a clear hierarchy of suitability and providing actionable recommendations tailored to different research objectives.

The investigation confirms that several high-quality datasets meet the primary criteria. The **University of Rochester Multi-Modal Music Performance (URMP) dataset** stands out as arguably the most versatile and comprehensive candidate. It provides isolated violin tracks from ensemble pieces, recorded at 48kHz in a controlled anechoic environment, and is accompanied by rich annotations including musical scores (MIDI/PDF) and manually corrected note-level transcriptions [2] [9]. Its primary considerations are the need for potential noise reduction and the student-level performance quality [10] [30]. The **Violin Etudes dataset** is a second top-tier candidate, specifically designed for performance analysis with 27.8 hours of monophonic etude recordings at 44.1kHz, offering constrained F0 annotations and shared MIDI files [21] [38]. For specialized tasks, the **TU-NOTE VIOLIN SAMPLE LIBRARY** offers pristine 48kHz recordings of single notes and transitions in an anechoic chamber, annotated with precise segmentation data, though its restrictive CC BY-ND 4.0 license limits its use in developing new models [6]. Finally, the **KRAISLER dataset** provides high-quality 44.1kHz violin tracks from piano-violine duets with exceptional MIDI synchronization and multiple acoustic versions, making it ideal for realistic audio generation [1].

Datasets failing the 44kHz+ sample rate threshold were also identified as valuable secondary options. The **Medley-solos-DB** dataset, with its 44.1kHz clips, is suitable for instrument recognition tasks, while the **TinySOL** and **Good-Sounds** datasets offer homogenous, high-quality recordings of single notes and exercises, respectively, which are useful for timbre modeling and foundational synthesis research [11] [15] [33].

Based on this comprehensive analysis, the following recommendations are provided:

1. **Prioritize Primary Candidates Based on Application:**

- For tasks involving **score following, source separation, or generating realistic solo violin in an orchestral context**, the **URMP dataset** is the recommended starting point due to its anechoic, isolated tracks and rich synchronization data [2].
- For tasks centered on **performance analysis, fundamental frequency (F0) estimation, vibrato modeling, or expressive timing**, the **Violin Etudes dataset** is the most appropriate choice, given its specialized design and F0 annotations [14] [35].
- For **timbral synthesis, physical modeling, or analysis of acoustic events like attacks and decays**, the **TU-NOTE VIOLIN SAMPLE LIBRARY** is the ideal resource, provided the project can accommodate its restrictive license [6].

1. **Evaluate Secondary Options for Specific Needs:**

- If the 44kHz+ requirement is flexible and the task is **instrument recognition or classification**, the **Medley-solos-DB** dataset is a readily available option, though its short clip length is a significant limitation [33].
- If the research focuses on **fundamental tone quality and basic timbre**, the **TinySOL** or **Good-Sounds** datasets provide clean, homogenous recordings that are easy to work with [11] [15].

1. **Conduct Due Diligence Before Commitment:**

- Before finalizing a dataset, it is strongly advised to download a small subset of the audio and perform a critical listening test. This will help identify any imperfections, such as the low-frequency noise reported in the URMP dataset, which may require pre-processing [30].
- Always verify the specific license terms. The difference between a permissive CC BY 4.0 license and a restrictive CC BY-ND 4.0 license can determine whether a dataset is usable for a particular research goal [6].

In conclusion, the search for high-quality, open-source monophonic solo violin data yields several excellent candidates. The optimal choice depends entirely on the specific requirements of the downstream task. By carefully considering the trade-offs between recording environment, annotation richness, content type, and licensing, a researcher can select the most suitable dataset to power their audio analysis or machine learning application.

Reference

1. KRAISLER: A Multi-Track Dataset of Piano and Violin Duet ... https://transactions.ismir.net/articles/10.5334/tismir.338
2. Creating A Multi-track Classical Music Performance ... https://arxiv.org/html/1612.08727v3
3. Automatic Assessment of Tone Quality in Violin Music ... https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2019.00334/full
4. AMAAI-Lab/ai-audio-datasets-list https://github.com/AMAAI-Lab/ai-audio-datasets-list
5. salu133445/bach-violin-dataset https://github.com/salu133445/bach-violin-dataset
6. (PDF) TU-Note Violin Sample Library – A Database of ... https://www.researchgate.net/publication/327704544_TU-Note_Violin_Sample_Library_-_A_Database_of_Violin_Sounds_with_Segmentation_Ground_Truth
7. URMP Dataset - University Lab Sites https://labsites.rochester.edu/air/projects/URMP.html
8. URMP Dataset: Multi-Modal Music Performance Corpus https://www.emergentmind.com/topics/university-of-rochester-multi-modal-music-performance-urmp-dataset
9. Datasets https://hackmd.io/@ben2002chou/rJcMmZcJp
10. "MIDI-DDSP: Detailed Control of Musical Performance via ... https://midi-ddsp.github.io/
11. TinySOL: an audio dataset of isolated musical notes https://zenodo.org/records/3685367
12. ViolinDiff, AI, and a possibile open source future https://linuxmusicians.com/viewtopic.php?t=28542
13. Contents | Bach Violin Dataset https://salu133445.github.io/bach-violin-dataset/contents.html
14. VIOLIN ETUDES: A COMPREHENSIVE DATASET FOR F0 ... https://repositori.upf.edu/bitstream/handle/10230/54154/Tamer_ismir_viol.pdf?sequence=1&isAllowed=y
15. Good-sounds.org dataset - MTG - Music Technology Group https://www.upf.edu/web/mtg/good-sounds

16. HIGH-RESOLUTION VIOLIN TRANSCRIPTION USING ... https://archives.ismir.net/ismir2023/paper/000025.pdf
17. Sanidha | Computational and Cognitive Musicology Lab https://ccml.gtcmt.gatech.edu/data/Sanidha/
18. Solos: A Dataset for Audio-Visual Music Analysis https://hackernoon.com/solos-a-dataset-for-audio-visual-music-analysis-related-work
19. From Waveforms to Instruments: Building a Convolutional ... https://medium.com/@patelpeeyush2000/from-waveforms-to-instruments-building-a-convolutional-audio-classifier-with-musicnet-d5553e3fa77a
20. Violin etudes: a comprehensive dataset for f0 estimation ... https://recercat.cat/handle/10230/54154
21. High-Resolution Violin Transcription using Weak Labels https://github.com/MTG/violin-transcription/
22. Leveraging Generative Models for Music and Signal ... https://johnthickstun.com/assets/pdf/thickstun_thesis.pdf
23. Score-informed Music Source Separation: Improving Synthetic-to ... https://arxiv.org/pdf/2503.07352
24. PHENICX-Anechoic: note annotations for Aalto ... https://www.upf.edu/web/mtg/phenicx-anechoic
25. note annotations for Aalto anechoic orchestral database https://zenodo.org/records/1289821
26. Anechoic Audio — ORASE https://www.orase.org/anechoic-audio
27. (PDF) Anechoic Recording System for Symphony Orchestra https://www.researchgate.net/publication/233609590_Anechoic_Recording_System_for_Symphony_Orchestra
28. Multi-Channel Orchestral Anechoic Recordings for ... https://www.semanticscholar.org/paper/Multi-Channel-Orchestral-Anechoic-Recordings-for/ea3e2907869037b84d9308c2237e670a399889b3
29. Anechoic audio and 3D-video content database of small ... https://hal.science/hal-02354814v1/file/000471.pdf
30. Improving Synthetic-to-real Generalization in Classical Music https://arxiv.org/html/2503.07352v1
31. HIGH-RESOLUTION VIOLIN TRANSCRIPTION USING ... https://repositori.upf.edu/bitstream/handle/10230/58121/Tamer_ism_high.pdf?sequence=1&isAllowed=y
32. Video Object Segmentation-Aware Audio Generation https://link.springer.com/article/10.1007/s11263-026-02911-2

33. Medley-solos-DB: a cross-collection dataset for musical ... https://zenodo.org/records/3464194
34. Hi-Fi, Studio-Grade Datasets https://www.useocular.com/human-data/hi-fi
35. nctamer/violin-etudes https://github.com/nctamer/violin-etudes
36. README.md - LAION-AI/audio-dataset https://github.com/LAION-AI/audio-dataset/blob/main/data_preprocess/README.md
37. ViolinEtudes https://zenodo.org/records/6564408
38. Violin Etudes: A Comprehensive Dataset for f0 ... - Zenodo https://zenodo.org/records/7316714
39. GitHub https://github.com/nctamer/violin-etudes/blob/328561def3e1b789670f541a76a99b5aa4841f9c/README.md